

Object Detection Metrics & Post-processing

IoU, NMS, and mAP

Goals

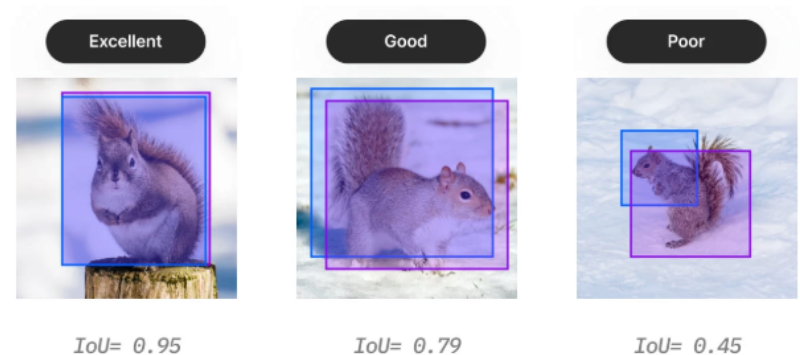
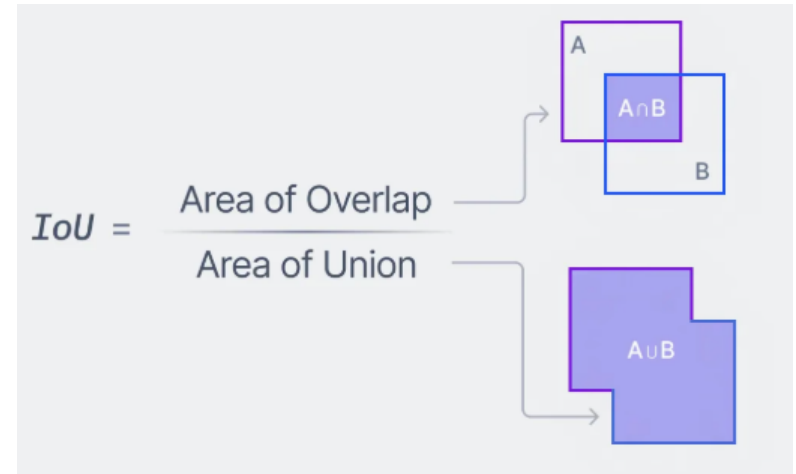
- Understand how detections match to ground truth via IoU
- Know how overlapping predictions are pruned with NMS
- Interpret precision, recall, AP, and mAP; avoid pitfalls

Detector Outputs Refresher

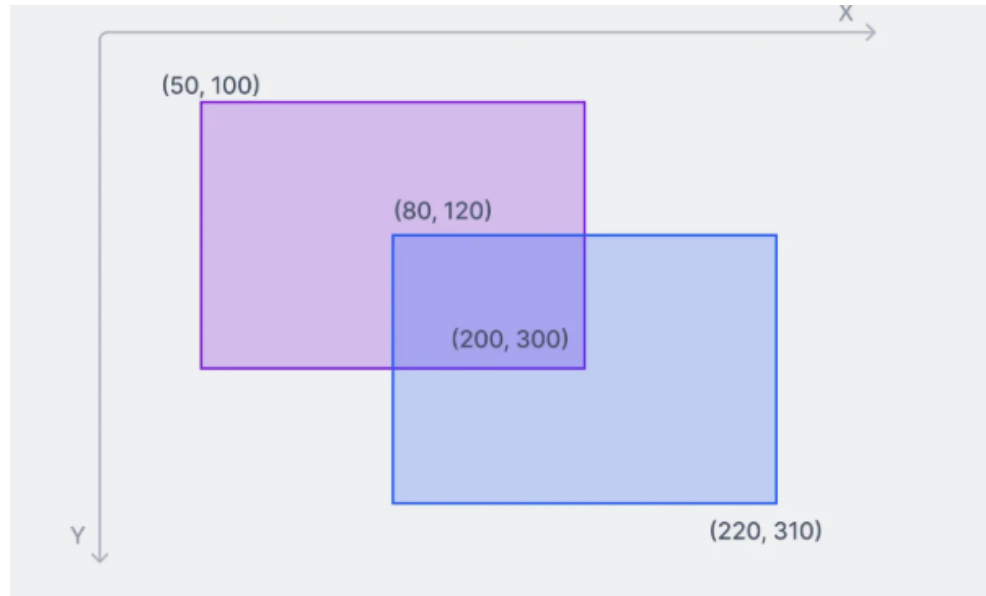
- For each image returns: boxes (x1,y1,x2,y2), class scores, confidence/objectness
- Multiple detections may target the same object → need post-processing
- Evaluation ties predictions to ground truth using IoU + class match

Intersection over Union (IoU)

- Measures quality between predicted and ground-truth boxes
- Range [0,1]; higher is better
- Used for matching: **TP** if class is correct **AND** $\text{IoU} \geq \text{threshold}$ (e.g., 0.5)



IoU example



Intersection = $(200 - 80) \times (300 - 120) = 21,600$

Purple = $(200 - 50) \times (300 - 100) = 30,000$

Blue = $(220 - 80) \times (310 - 120) = 26,600$

$IoU = 21,600 / 56,600 = 0.62$

Non-Max Suppression (NMS) — Why and How

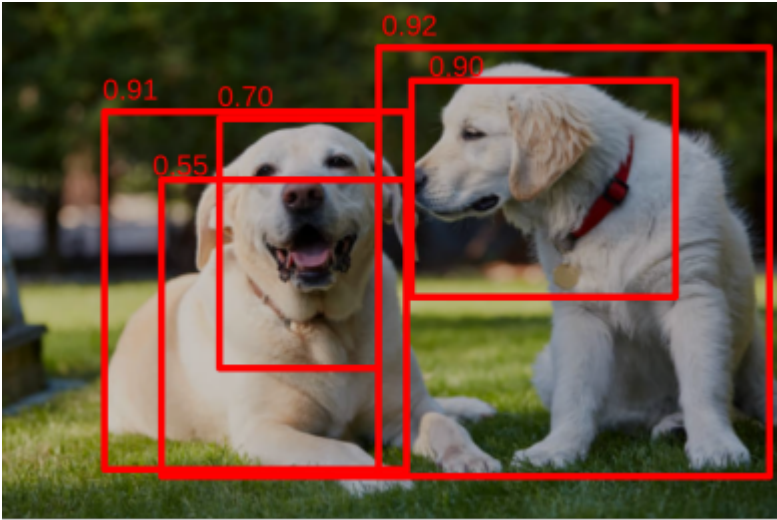
- Why: detectors output many overlapping boxes per object
- NMS: sort boxes by score, keep best, drop overlaps above IoU_{th}

NMS example



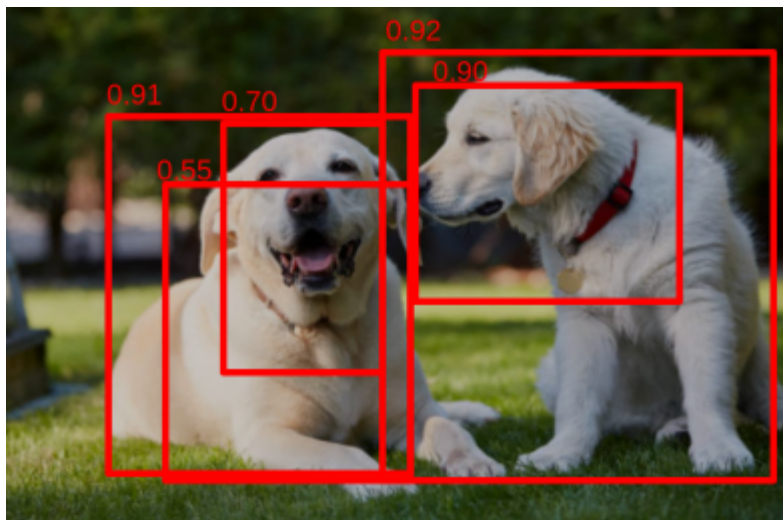
For this Image

NMS example



Object detector proposed these bounding boxes and confidences.

NMS example



Algorithm

- Save highest confidence box
- Compute IOU with all other boxes
- Delete bounding boxes with IOU > threshold (say 0.2)

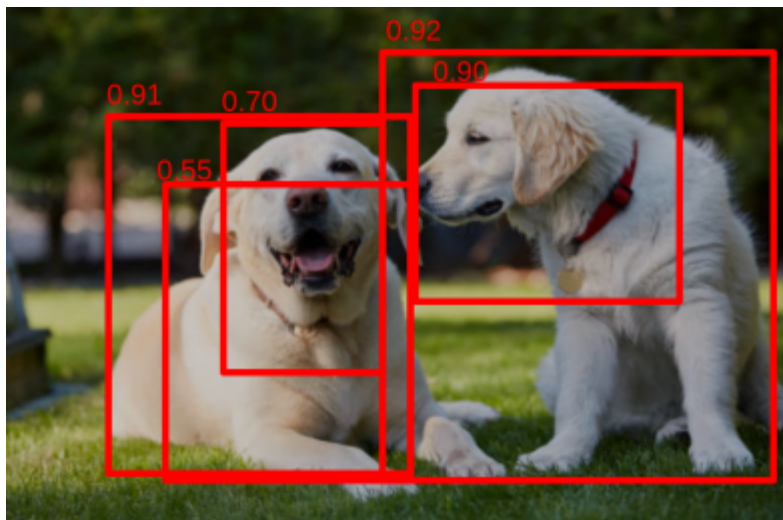
Proposals

Box #	Score	IOU
1	.91	
2	.70	
3	.55	
4	.92	
5	.90	

Keepers

Box #	Score

NMS example



Algorithm

- Save highest confidence box
- Compute IOU with all other boxes
- Delete bounding boxes with IOU > threshold (say 0.2)

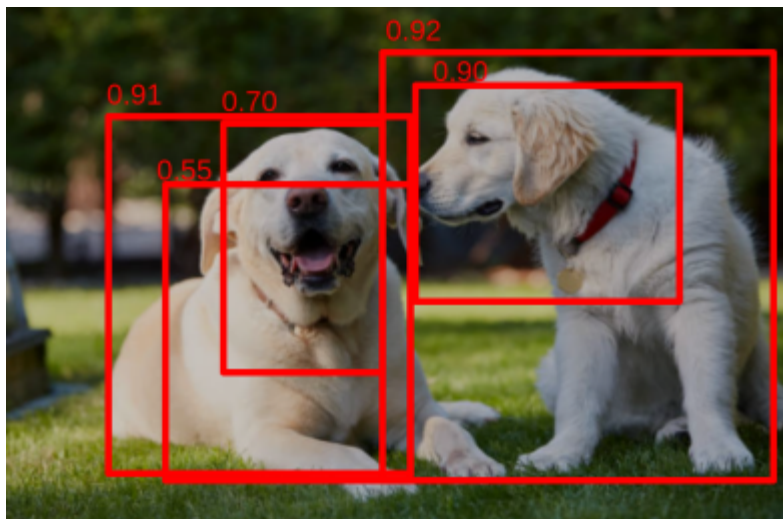
Proposals

Box #	Score	IOU
1	.91	
2	.70	
3	.55	
5	.90	

Keepers

Box #	Score
4	.92

NMS example



Algorithm

- Save highest confidence box
- **Compute IOU with all other boxes**
- Delete bounding boxes with IOU > threshold (say 0.2)

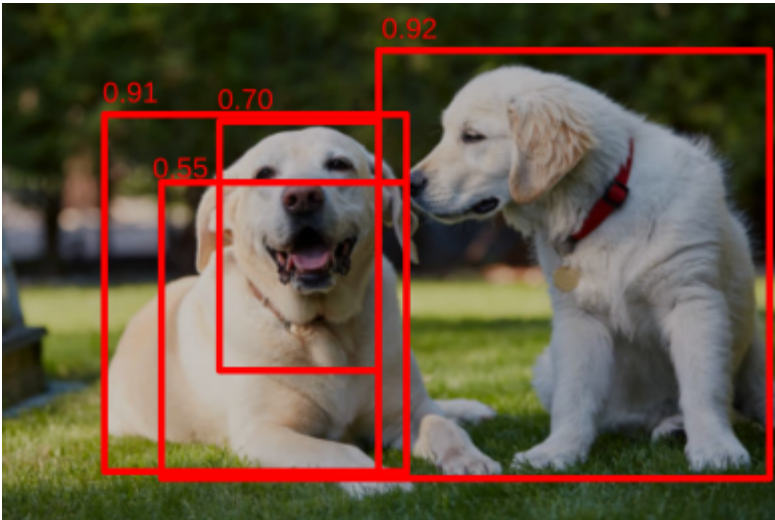
Proposals

Box #	Score	IOU
1	.91	0
2	.70	0
3	.55	0
5	.90	0.3

Keepers

Box #	Score
4	.92

NMS example



Algorithm

- Save highest confidence box
- Compute IOU with all other boxes
- Delete bounding boxes with IOU > threshold (say 0.2)

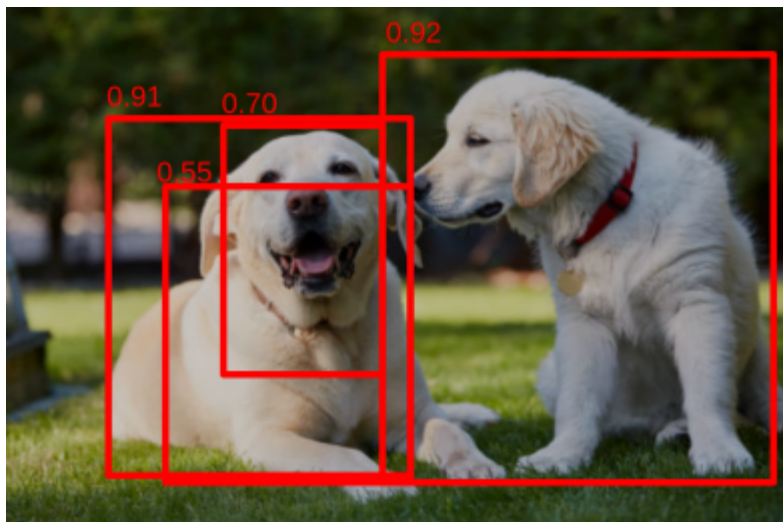
Proposals

Box #	Score	IOU
1	.91	0
2	.70	0
3	.55	0

Keepers

Box #	Score
4	.92

NMS example



Algorithm

- Save highest confidence box
- Compute IOU with all other boxes
- Delete bounding boxes with IOU > threshold (say 0.2)

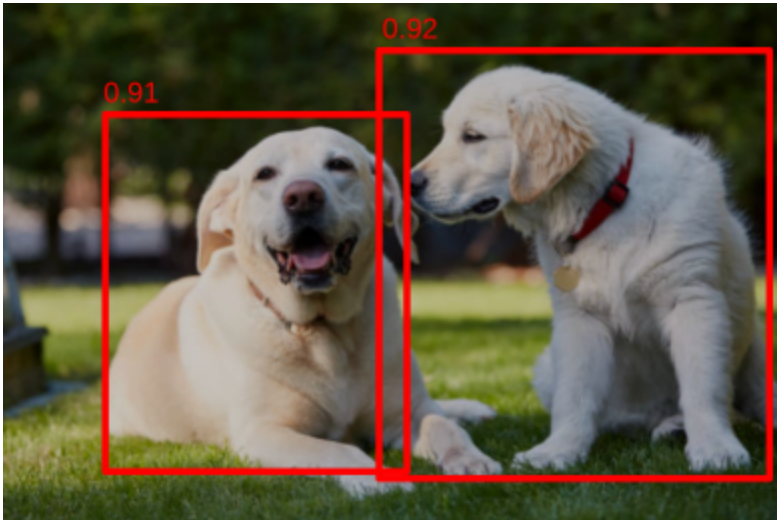
Proposals

Box #	Score	IOU
2	.70	.3
3	.55	.7

Keepers

Box #	Score
4	.92
1	.91

NMS example



Algorithm

- Save highest confidence box
- Compute IOU with all other boxes
- Delete bounding boxes with IOU > threshold (say 0.2)

Proposals

Box # Score IOU

Keepers

Box # Score

4	.92
1	.91

Precision, Recall

- Precision = $TP/(TP+FP)$ → of what you predicted, how much was correct?
- Recall = $TP/(TP+FN)$ → of what exists, how much did you find?

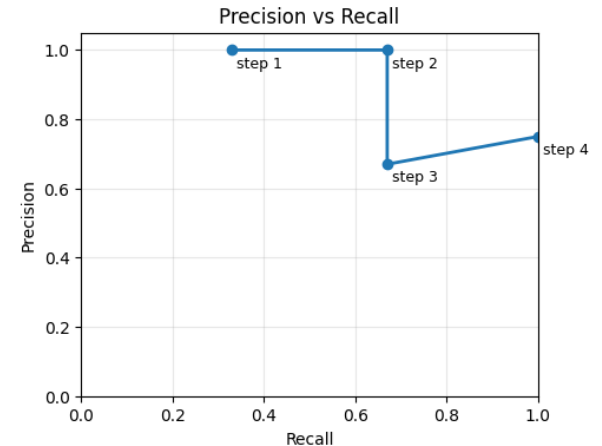
AP

- Measures how good a detector is for one class by summarizing its whole precision–recall trade-off into a single number. Fix an IoU threshold (e.g., 0.5), **sweep the score/confidence threshold from high→low**, plot precision vs. recall, and take the area under that curve. Bigger area $\Rightarrow$ better detector for that class.

AP

- Ex. Suppose one Class A has 3 ground-truth objects. And you are using a IoU threshold of .5. As you lower the model score threshold (confidence) you accumulate predictions:

step	TP	FP	Precision	Recall
1	1	0	1.00	0.33
2	2	0	1.00	0.67
3	2	1	0.67	0.67
4	3	1	0.75	1.00



Take the area under the curve of plot (max of $1 \times 1 = 1$). Say it comes out to 0.83

That means for this class you have an $AP@0.5 = .83$

mAP

mAP stands for **Mean Average Precision** and is the most common metric used to evaluate the performance of object detection models in computer vision. It provides a single number that summarizes a model's performance across all object classes and at different levels of detection quality.

A high mAP score indicates the model is good at two things:

1. **Correctly identifying** the object's class (High **Precision**).
2. **Accurately locating** the object in the image (High **Recall** and accurate bounding box).

mAP

Ex. Two classes – calculate mAP@0.5

Class A AP@0.5 = 0.83 (from previous slide)

Class B (different predictions/ground truths) AP@0.5 = 0.60

Then mAP@0.5 across these two classes is the mean:

$$\text{mAP@0.5} = \frac{0.83 + 0.60}{2} = 0.715$$